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Stock-market disruptions and corporate disclosure policies☆

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ABSTRACT

We examine managers'' disclosure decisions in response to non-fundamental price shocks. Using mutual fund fire sales as a source of market disruption, we show some firms respond by issuing earnings guidance. Others, especially firms with weaker performance and more short-term-oriented investors, engage in accrual-based earnings management. To identify the efficacy of firm disclosure choices, we examine passage of Sarbanes-Oxley and Regulation Fair Disclosure and show that they increased reliance on guidance rather than earnings management. The shift is associated with faster post-fire-sales price recovery, suggesting enhancing information disclosure rather than manipulation is effective in mitigating the effect of market disruptions.

1. Introduction

Recent studies show that disruptions in secondary stock markets can adversely impact firms' real economic decisions and performance. For example, stock mispricing can impede firms from raising equity capital and distort their investments (Lou and Wang (2018); Khan et al. (2012); Hau and Lai (2013)). Liquidity shocks to stock prices can also create negative externalities by conveying unfavorable signals to stakeholders, such as customers and suppliers (Subrahmanyam and Titman (2001); Williams and Xiao (2017); Dessaint et al. (2019)). Moreover, firms with underpriced stocks are more likely to become takeover targets, posing a significant threat to incumbent management (Edmans et al. (2012)). To the extent managers perceive a stock price drop to be unrelated to firm fundamentals, we would expect them to take deliberate steps to counter the price decline. In this study, we investigate managerial response – in the form of voluntary disclosure and financial reporting – to stock market disruptions, and the effectiveness of these actions in facilitating a recovery from these liquidity disruptions.

We use mutual fund flow-driven fire sales (Coval and Stafford (2007); Edmans et al., 2012; Wardlaw (2020)) as a source of exogenous disruption to stock prices. As Coval and Stafford (2007) show, mutual funds that are subject to extreme capital outflows create price pressure when they are forced to sell stocks they hold in common. This price pressure is unrelated to firm fundamentals, but can induce prolonged mispricing that, on average, takes more than a year to recover (Edmans et al. (2012)). Consistent with the

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literature, we find that mutual fund outflow-driven price pressure can trigger substantial price drops. There is, however, considerable variation in the price impact across different stocks: Our cross-sectional tests show that firms with a better information environment, as reflected by higher analyst forecast quality and stock liquidity, experience less negative impacts when their stocks are subject to mutual fund fire sales.

We investigate firms' response to mutual fund fire sales in terms of their financial reporting and voluntary disclosure policies. Our hypothesis is that increased voluntary disclosure may enable firms to improve the information environment and hence mitigate the adverse price impact of fund flow-driven fire sales. Our tests reveal that firms tend to increase the frequency of earnings forecasts in response to mutual fund fire sales. Furthermore, these firms issue management earnings forecasts with greater precision, i.e., forecasts with point estimates as opposed to range estimates. This is consistent with our hypothesis that firms use voluntary disclosure to improve transparency (Diamond and Verrecchia (1991); Leuz and Verrecchia (2000)). The effort by firms to provide informative earnings guidance delivers benefits: we find that post-fire-sales management earnings forecasts, especially those with lower error, are associated with moderating the negative price impact of fire sales and a faster price recovery.

Despite the above evidence, it is not clear whether all firms under the mutual fund flow-induced price pressure would choose earnings guidance as a response. For instance, firms with weak operating performance may not be in a position to disclose information that is sufficiently positive to counter the price impact of mutual fund fire sales. We examine the cross-sectional variation in firms' disclosure policies after mutual fund fire sales, and find that firms with lower pre-fire-sales operating performance do not increase earnings guidance frequency in response to fire sales. Instead of disclosing information as a response, these firms use discretionary accruals to manipulate accounting earnings upward. While the use of earnings management is compatible with managers' incentives to counter the negative price impact of fire sales, the implications are antithetical in terms of the information environment – earnings management tends to induce informational opacity rather than transparency (Sloan (1996)). Hence, we look further into firms' use of earnings management as a response to mutual fund fire sales and its effectiveness in mitigating the negative price impact.

As Coval and Stafford (2007) show, market disruption due to mutual fund fire sales appears to be corrected by the market over time. We would, however, expect managers that are more concerned about current stock price levels to be more proactive in responding to mutual fund fire sales. Managers' focus on short-term price movements may be keener in the presence of shorter-term investors. We therefore examine the cross-sectional variation in investor horizons, reflected by investors' turnover rate of their portfolios (Gaspar et al. (2005)), institutional ownership (Aghion et al. (2013)), and presence of blockholders (Edmans (2009)). We find that firms significantly increase discretionary accruals after mutual fund fire sales when investors' turnover rate is high, the institutional ownership is low, or when blockholders are absent. These represent situations where the average investor horizon is short. On the contrary, firms with relatively long-term oriented investors tend to rely on earnings guidance rather than earnings management as a response to the non-fundamental price shocks. Taken together, this evidence indicates that firms with managers that are particularly concerned about short-term price levels tend to use earnings management in response to mutual fund fire sales.

Though some managers respond to mutual fund fire sales by increasing earnings management, we would not generally expect such a response to be effective at mitigating the price impact of fire sales. First, as we argue above, informational transparency rather than opacity is more likely to help in correcting mispricing. Second, our findings suggest that firms that use earnings management may also have lower pre-fire-sales firm performance. Thus, it is possible that lower stock prices following mutual fund fire sales may partly reflect the fundamental value of these firms, rather than mispricing as such. Consistent with predictions, we find that post-fire-sales discretionary accruals are not associated with mitigating the price impact of fire sales or leading to faster price recovery.[1]

However, since firms' responses appear to be endogenous to firm characteristics, we cannot make any causal claims about the effect of disclosure policies on price recovery on the basis of these results. For instance, since firms' disclosure responses are related to firm performance, it is possible that better-performing firms will have a faster recovery from fire sales regardless of actions taken by managers. To address the endogeneity in firms' disclosure policy responses to mutual fund fire sales, we make use of two quasi-natural experiments: Regulation Fair Disclosure of 2000 (Regulation FD) and the Sarbanes-Oxley Act of 2002 (SOX). The extant literature shows that Regulation FD, that required material information to be shared with all investors on an equal footing, was followed by a substantial increase in voluntary earnings forecasts (Heflin et al. (2003)). Existing studies also show that the passage of SOX and its enforcement of board independence deter the use of discretionary accruals (Cohen et al. (2008)) and enhance corporate transparency (Armstrong et al. (2014)). Thus, the passage of Regulation FD and SOX might force firms to shift from earnings management to earnings guidance as a response to mutual fund fire sales. Using a difference-in-differences test around the passage of Regulation FD and SOX, we find that firms are indeed more (less) likely to use earnings guidance (discretionary accruals) in response to mutual fund fire sales after the regulatory changes. Further, these regulatory changes are associated with faster price reversals after fire sales. These results further suggest that enhanced information disclosure rather than earnings management helps mitigate the adverse effect of mutual fund fire sales. Strengthening the causal interpretation of our results, we find that the changes in disclosure policies and speed of price recovery were concentrated among firms that were impacted by SOX passage. These are the firms that were not compliant with SOX requirements prior to 2002, i.e., did not have a majority of independent directors and a fully independent audit committee prior to

[1] There are three possible explanations for why managers use earnings management in response to price shocks. First, as discussed above, firms with stocks that underperform pre-fire-sales and managers that are more concerned about short-term price movements may not be in a position to disclose sufficiently positive information to counter the exogenous price drops. Second, managers may mistakenly believe this action to be helpful in reversing the stock price decline. Third, even if managers do not expect earnings management to be useful in facilitating price recovery, they may be compelled to do it if investors expect earnings management in equilibrium and rationally discount reported earnings for some level of anticipated manipulation. In this case, a firm that did not engage in earnings management could be regarded by investors as being worse than it actually was.

SOX.

Our study contributes to three strands of literature. First, we provide evidence on the causal effect of stock prices on managerial disclosure decisions. Although the literature has long recognized that maintaining or increasing stock prices is one of the most important factors for managers' reporting and disclosure decisions, the extant literature mainly examines the reverse causality — i.e., the effect of voluntary earnings forecasts and earnings management on stock prices.[2] Several recent studies have attempted to examine the effect of stock prices on voluntary disclosure policies. For example, Sletten (2012) shows that managers respond to negative news of their industry peers by engaging in strategic voluntary disclosure as a way to mitigate negative price impacts on their firms. Li and Zhang (2015) exploit the adoption of Regulation SHO as a shock to short selling activities and find that firms tend to respond by reducing the precision of bad news forecasts. Balakrishnan et al. (2014) find that firms disclose information voluntarily when they suffer exogenous shocks to their public information supply, and such responses are shown to effectively improve liquidity. Kadach (2016) suggests that managers are less likely to issue corrective earnings forecasts if they exploit option exercise strategies during mispricing. Zuo (2016) documents that managers incorporate private information into stock prices in their earnings forecasts. All these studies examine the response of voluntary disclosure to (negative) stock price information. Our paper differs from these studies in that mutual fund fire sales reduce both price *levels* and price *efficiency*, because these trades are driven by liquidity shocks to mutual funds rather than information. Moreover, our paper provides evidence on how voluntary disclosure and financial reporting shape the dynamic pattern of mispricing, by investigating not only how managers respond to exogenous price shocks but also to establish how their responses causally influence price reversals.

Second, our study sheds light on the underlying incentives behind managers' choice of disclosure policy changes in response to exogenous shocks: while firms on average tend to respond to downward price pressure with information disclosure, firms that do not have positive information to disclose or are concerned about short-term firm value, respond by manipulating accounting earnings. These findings are related to the vast literature on the determinants of managerial decisions on financial reporting and disclosure policies (e.g., Verrecchia (1990); Leuz et al. (2003); Bergstresser and Philippon (2006); Hazarika et al. (2012); Kedia et al. (2015); Fang et al. (2016)), by recognizing that managers also respond to non-fundamental shocks to stock prices in dissimilar ways due to different performance or underlying incentives.

Third, our findings on managerial responses to market disruptions add to the growing literature on the link between financial markets and real economic activities (Bond et al. (2012)). Recent studies show that exogenous stock mispricing due to mutual fund fire sales/purchases can constrain firms' financing and hence distort investment decisions (Khan et al. (2012); Hau and Lai (2013); Lou and Wang (2018)). Given these real effects of stock mispricing, firm managers are expected to deliberately mitigate stock mispricing as long as they recognize it. Our evidence shows that these firms indeed adjust their financial reporting and voluntary disclosure policies to mitigate exogenous stock price shocks.

The remainder of the paper is organized as follows. We review the literature and develop our hypotheses in the next section. Section 3 describes the data and variable construction. Section 4 discusses our empirical approach and presents the results. Section 5 concludes.

2. Hypothesis development

Existing studies have shown that market disruption can induce various adverse consequences, such as constraining firms from raising capital (Lou and Wang (2018); Khan et al. (2012); Hau and Lai (2013)), conveying false negative signals to stakeholders (Subrahmanyam and Titman (2001); Williams and Xiao (2017)), and increasing takeover risk (Edmans et al. (2012)). When mutual fund fire sales take place, managers may not be fully aware of the reasons for the stock price decline. However, as long as they believe that the price decline is not driven by firm fundamentals, we expect them to respond by adjusting their disclosure policies, in the hope of reversing the price decline as well as preventing the adverse consequences.

Management earnings forecasts are an important source of information, accounting for over 15% of the quarterly return variance and approximately 55% of accounting-based information (Hirst et al. (2008); Beyer et al. (2010)). An extensive prior literature suggests that voluntary disclosure is associated with lower information asymmetry. Theoretical models provide persuasive arguments that a commitment to higher levels of disclosure reduces information asymmetries arising either between the firm and its shareholders or among investors. This, in turn, should lower the cost of raising capital (see, e.g., Diamond and Verrecchia (1991); Baiman and Verrecchia (1996)). Consistent with this notion, Coller and Yohn (1997) and Leuz and Verrecchia (2000) show that both the issuance of management earnings forecasts and the strengthening of financial disclosure requirements are associated with higher stock liquidity. A more recent literature stresses the important role of voluntary disclosure in supplementing information, especially when public information is deficient or obfuscatory. Balakrishnan et al. (2014) show that managers respond to exogenous decreases in public information by increasing voluntary disclosure. Further, this increase in disclosure is shown to have causal effects on liquidity and cost of capital. Hence, when mutual fund fire sales occur, we would expect firms to respond by disclosing more information. Our hypothesis can be stated as:

H1: Firms are likely to increase issuance of voluntary earnings forecasts in response to mutual fund fire sales.

We also consider the possibility that some firms respond to mutual fund fire sales by manipulating their accounting earnings to convey false positive information. The literature suggests that firms manage earnings using discretionary accruals to meet earnings

[2] See, e.g., Pownall et al. (1993); Coller and Yohn (1997); Teoh et al. (1998a); Teoh et al. (1998b); Anilowski et al. (2007); Rogers et al. (2009).

targets such as analysts' forecasts (Burgstahler and Dichev (1997); Matsumoto (2002)). Various papers find opportunistic use of discretionary accruals, particularly around corporate events, such as IPOs and SEOs (Rangan (1998); Teoh et al. (1998a); Teoh et al. (1998b); Kim and Park (2005)), debt covenant violations (DeFond and Jiambalvo (1994)), and insider trading (Darrough and Rangan (2005); Agrawal and Cooper (2015)). In the case of mutual fund fire sales, we hypothesize that the stock price decline will pressure some managers to engage in accrual-based earnings management.

Earnings forecasts and earnings management have opposing effects on the information environment, because earnings management tends to induce informational opacity rather than transparency. It is unlikely, therefore, that the same group of firms will adopt both strategies in response to non-fundamental price declines. Consistent with this view, Francis et al. (2008) show that firms with better (worse) earnings quality have more (less) voluntary disclosure. Studies have also shown that firms' use of earnings guidance is positively related to operating performance (e.g., Miller (2002); Houston et al. (2010)). In the case of mutual fund fire sales, we might expect underperforming firms to be unwilling to issue earnings forecasts. Instead, these firms could adopt upward-biased discretionary accruals in response to non-fundamental price shocks. Hence:

H2: Firms with lower pre-fire-sales operating performance are likely to use discretionary accruals instead of voluntary earnings forecasts in response to mutual fund fire sales.

A number of empirical studies have documented the relation between institutional ownership and information disclosure. For example, Boone and White (2015) and Bird and Karolyi (2016) show that exogenous increases in passive institutional investor ownership are associated with more information disclosure from managers and improvements in the information environment. Shareholders' horizon is also likely to affect firms' response to mutual fund fire sales. Institutional investors, especially blockholders, can induce greater information disclosure because of their incentives to produce and trade on fundamental information (Edmans (2009)). Institutional investors have been shown to mitigate managerial myopia and deter managers from opportunistic actions such as earnings management (Bushee (1998); Jiambalvo et al. (2002); Aghion et al. (2013)). We infer shareholders' horizon based on the average turnover rate of the investors' portfolio (Gaspar et al. (2005)), the level of institutional ownership, and the number of blockholders. We hypothesize that firms with more shorter horizon investors are more likely to take deliberate steps in response to mutual fund fire sales. These firms are more likely to use earnings management as a response when they are underperforming and unwilling to convey true signals to the market.

H3: Firms with shorter-horizon investors are more (less) likely to use discretionary accruals (management earnings forecasts) in response to mutual fund fire sales pressure.

We next discuss the effectiveness of managerial responses to mutual fund fire sales. Existing studies show that managerial earnings forecasts have information content that affects stock prices (e.g., Pownall et al. (1993); Anilowski et al. (2007)) and improves the information environments (Coller and Yohn (1997); Balakrishnan et al. (2014); Boone and White (2015); Bird and Karolyi (2016)). On the other hand, earnings management is generally regarded as a cause of stock mispricing. For example, Teoh et al. (1998a) and Teoh et al. (1998b) find that IPO and SEO firms with high accruals have worse abnormal returns after equity issuance. Xie (2001) shows that abnormal accruals are associated with stock overpricing. Chan et al. (2006) show that high accruals are associated with lower future stock returns.

In a mutual fund fire sale, trading by funds is driven by their need for liquidity. It is akin to noise trading (i.e., non-information driven trading) and can be expected to introduce considerable noise into the price of the affected stock. Such noise trading may increase investor uncertainty about the economic fundamentals of the firm and the reliability of the information that is available. As a result, investors may be inhibited from producing and trading on information (De Long et al. (1990); Shleifer and Vishny (1997); Dow et al. (2016)).[3] Information disclosure from insiders can thus reduce information uncertainty, encouraging investors to collect and trade on information (since this could lower the effective cost of collecting additional information), thereby alleviating mispricing (Kyle (1985)). Hence, we predict that managerial actions such as earnings forecasts, that improve informational transparency, will be more effective in mitigating the impact of mutual fund fire sales.

H4: Management earnings forecasts, especially those with higher quality, are more effective in mitigating the price impact of mutual fund fire sales compared to earnings management.

Finally, since the choice of disclosure strategy – earnings management versus earnings guidance – is likely to be affected by firms' operating performance, we cannot give a causal interpretation to the association between strategy choice and price recovery following a fund flow-driven fire sale. To identify the effect of disclosure strategy, we rely on two quasi-natural experiments, the passage of Regulation FD and SOX, that have been shown to encourage earnings guidance and discourage earnings management, respectively. We expect that when firms encounter the non-fundamental price declines, the passage of Regulation FD induces greater disclosure, and SOX reduces earnings management. As a result, both regulatory changes facilitate price recovery from fire sales. Besides, we expect firms that are affected by SOX requirements ('non-compliant' firms) to exhibit a clear pattern of not relying on earnings management post-SOX. Therefore, non-compliant firms are expected to exhibit a more rapid price recovery post-SOX, relative to firms unaffected by SOX passage.

H5: We expect the greater (less) use of earnings guidance (earnings management) following the passage of Regulation FD (SOX) to facilitate price recovery following mutual fund fire sales. For SOX, the effect is expected to be concentrated among firms that were not compliant with SOX requirements prior to its passage.

[3] For instance, investors may regard the greater uncertainty as an escalation in the cost of gathering reliable information and, thereby, choose to not gather additional information.

3. Data and measurement of main variables

3.1. Data

We begin with all U.S. public firms with financial information from the intersection of CRSP and Compustat. We obtain data on management earnings forecasts and institutional ownership from I/B/E/S Guidance and Thomson Reuters, respectively. Our sample covers firm-quarter observations from 1994 to 2016.[4] We exclude the following firm-quarter observations: (1) firms with missing accounting or stock price information; (2) financial institutions and insurance companies (4-digit SIC codes 6000–6999); (3) firms in regulated industries (4-digit SIC codes 4900–4999); (4) firms with a market value of equity less than 5 million dollars. Our final sample consists of 416,849 firm-quarter observations.

3.2. Managerial responses

The two main dependent variables in our study are *Freq_Guidance* and *DisAccrual*. *Freq_Guidance* is the natural logarithm of one plus the number of quarterly earnings forecasts issued by the management during a fiscal quarter. *DisAccrual* is discretionary accruals measured using modified Jones Model following the earnings management literature (Dechow et al. (1995)). We also look into management earnings forecast characteristics, including *Forecast News*, *Guidance Precision*, and *Guidance Error*. Following Cheng et al. (2013) and Li and Zhang (2015), we measure *Forecast News* as the difference between the point estimate (or the midpoint estimate of a range forecast) and the consensus analyst forecast, scaled by beginning-of-quarter stock price. The consensus analyst forecast is the median of analyst forecasts at the time of the management earnings guidance. Based on the sign of *Forecast News*, each management earnings forecast is classified as positive (*Forecast News* > 0) or negative (*Forecast News* < 0) forecast news. We code *Bad News* as one if a firm issues more negative forecast news than positive news over a quarter, and zero otherwise. *Guidance Precision* is a dummy variable set equal to one if a firm issues more point forecasts than range forecasts over a quarter, and zero otherwise. *Forecast Error* is defined as the absolute difference between a management earnings forecast and the actual earnings, scaled by beginning-of-quarter stock price.[5]

3.3. Mutual fund flow-driven price pressure

We follow the literature (see, e.g., Coval and Stafford (2007); Edmans et al. (2012); Wardlaw (2020)) and construct mutual fund fire sales as an exogenous shock to stock prices. We collect data on mutual fund holdings from Thomson Reuters and mutual fund returns from CRSP, and remove funds that specialize in a particular industry to address the concern that mutual fund flows might be driven by industry fundamentals. We construct the hypothetical mutual fund flow-driven trading measure as follows:

$$MFFlow_{i,t} = \frac{S'_{i,t-1}F_t}{v_{i,t}},$$

where $S_{i,\,t-1}{}' = (s_{i,\,1,\,t-1}, \ldots, s_{i,\,j,\,t-1})$ is a $1 \times J$ vector of the number of stock $i's$ shares owned by mutual fund j at the beginning of quarter t, $F_t' = (f_{1,\,t}, \ldots, f_{j,\,t})$ is a $1 \times J$ vector of fund $j's$ percentage outflow over quarter t, and $v_{i,\,t}$ is total share trading volumne of stock i in quarter t. The measure is summed over mutual funds that have outflows equal to or greater than 5% to ensure that the measure captures mutual fund trading driven by liquidity shocks.[6] We winsorize $f_{j,\,t}$ at 90% to ensure that the measure is not driven by spurious data. We take the natural logarithm of the absolute value of the magnitude of the shock measure:

$$MFS_{i,t} = Ln(1 + |MFFlow|),$$

where a higher value of *MFS* indicates a stronger price pressure from fire sales. We use *MFS* in most of our empirical analysis.

The exogeneity assumption is that price pressure created by large mutual fund outflows is not related to a firm's fundamentals, which is justified in Lou (2012). However, several recent studies raise concerns regarding the exogeneity of the shock measure. One concern is that even when mutual funds suffer from liquidity shocks, they can still trade based on information about the firm's economic prospects. However, as in Edmans et al. (2012), *MFFlow* is constructed using mutual fund holdings prior to the large outflows, and is thus unlikely to capture endogenous selection by these mutual funds (Berger (2019)). This concern is also alleviated to some extent by removing sector funds, so that fund flows and trades are unlikely to reflect industry-level information.[7]

[4] The sample starts in 1994 due to the availability of I/B/E/S Guidance. Although this data first became available in 1992, there were only 40 observations before 1994.

[5] It is also important to control for determinants of management earnings guidance, as suggested by Rogers and Van Buskirk (2013). The definitions are described in Section 4.2.

[6] The 5% threshold is proposed by Edmans et al. (2012). In our sample, the median quarterly fund flow is −0.3%, and −5% is below the 20th percentile of fund flow. Thus, −5% seems to be a reasonable cutoff that represents a sizable outflow while still includes a significant proportion of mutual funds.

[7] Wardlaw (2020) also shows that the original *MFFlow* measure developed by Edmans et al. (2012) overstates the price impact of mutual fund fire sales. Thus, we make the adjustment following Wardlaw (2020) by excluding stock price from the original *MFFlow* measure. Note, however, our findings remain similar with the original *MFFlow* measure.

Another concern is that fund flows can reflect fundamental information about the underlying stocks possessed by fund investors. For example, Cumming et al. (2019) use proprietary data from Canadian mutual funds to show that different types of fund flows, such as new purchases, one-time redemptions, and switch flows, can contain different amounts of information. Since data on types of fund flows is unavailable to us, we cannot rule out the possibility that our net fund flows are driven by more informative sources and our measure is endogenous to firm fundamentals. However, such concerns regarding our fund flow measure are mitigated since we focus on funds that experience an unusually large outflow ($< -5\%$). Such large outflows are more likely driven by liquidity shocks than information. In addition, as previous studies (e.g., Edmans et al. (2012); Heater et al. (2017)) as well as our Fig. 1 show, a large negative value of *MFFlow* is related to a temporary decline in the stock price that is eventually reversed, indicating that the price movement is driven by noise rather than fundamentals. Furthermore, in an unreported test, we find that our fund flow measure has no predictive power for the future operating performance of the underlying stocks. Therefore, while our measure might include informative fund flows such as those driven by switch-in/switch-out decisions by sophisticated investors, it seems to be primarily driven by uninformative flows that cannot predict future firm performance.[8]

3.4. Other explanatory variables

In the regression analysis, we control for the quality of firms' information environment and stock liquidity. *Analyst Coverage* is the number of analysts following a firm reported in I/B/E/S. We use two measures of stock liquidity: *Amihud's Illiquidity* and *Turnover*. We also include a set of control variables that are considered to be related to price efficiency and/or managers' decisions on earnings forecasts and earnings management. We first control for prior stock return characteristics, including buy-and-hold abnormal returns over the year prior to the event quarter (*BHAR*), average daily returns (*MRet*), standard deviation of monthly returns over the past year (*Sigma*), and stock trading turnover (*Turnover*), defined as the average daily trading volume divided by the number of total outstanding shares. In addition, we include the following firm characteristics: firm size (*Size*) is the natural logarithm of total assets at the end of last quarter; *Tobin's Q* is the sum of total assets and the difference between market value and book value of common equity, divided by total assets; financial leverage (*Leverage*) is the sum of long-term debt and debt in current liabilities divided by total assets; *ROA* is earnings before extraordinary items divided by lagged total assets; *MF Ownership* is the ratio of the number of shares held by mutual fund investors (included in the calculation of *MFFlow*) to the total number of shares outstanding for a firm; *Inst. Ownership* is the ratio of the number of shares held by institutional investors (excluding mutual funds in the calculation of *MFFlow*) to the total number of shares outstanding for a firm. More detailed definitions of our variables are provided in Appendix A.

3.5. Descriptive statistics

Panel A of Table 1 presents the summary statistics of the variables we use in the empirical analysis. We winsorize all the variables at the 1st and 99th percentiles. *MFFlow* is less than or equal to zero because it only captures outflow-driven mutual fund sales. The sample mean of *MFFlow* is -0.618, indicating that the average hypothetical flow-driven pressure amounts to 0.618% of the actual trading volume of the stock. The average of *Freq_Guidance* and *DisAccrual* are 0.091 and 0.082, respectively. The summary statistics for other variables are comparable with the extant literature.

Panel B of Table 1 shows the correlation matrix of the variables. The negative correlation between *MFS* and $CAR[0,0]$ (average monthly abnormal returns during the quarter of mutual fund fire sale) indicates that a greater pressure of mutual fund fire sales is related to a more negative concurrent stock return. *Freq_Guidance* is positively correlated with *Size*, *ROA*, and *Analyst*, while negatively correlated with *Amihud*. This evidence indicates that larger firms with better historical performance, more analyst followings, and higher stock liquidity tend to issue quarterly earnings forecasts more frequently. The correlation between *DisAccrual* and *ROA* is 0.172, which might be reflective of firms using discretionary accruals to inflate accounting earnings.[9]

4. Model specification and empirical results

4.1. The Price impact of mutual fund fire sales

We first graphically present the price impact of mutual fund fire sales in Fig. 1, which considers stocks with *MFFlow* ranked in the bottom decile of the quarter and plots the cumulative average abnormal returns over equal-weighted market return from 6 months before to 18 months after the fire sales. In this figure, the fire sale takes place from the beginning of month -2 to the end of month 0 (i.

[8] Again, the exogeneity assumption for the price shock measure is that it captures the projected price shocks due to large outflows from mutual funds that hold the same stock. And those distressed mutual funds will sell the underlying stocks proportional to the amount of holding. This assumption does not hold for inflow-induced purchases because mutual funds do not have to buy stocks they already own proportionally, even when they experience large inflows. In an untabulated test, we find that a similar measure using mutual funds with large inflows does not create a significantly positive impact on stock prices.

[9] As the correlation matrix indicates that there may be strong correlations among some of the variables, in unreported tests we estimate the variance inflation factors (VIF) for the main regressions as presented by Equations (3) and (6) to examine whether multicollinearity is a concern. Our estimates show that only two variables, *Size* and *Amihud*, have the VIF above the conventional critical value of 10. Also, our main findings remain similar after excluding these variables. Therefore, multicollinearity does not seem to be a severe problem in our estimation.

Table 1
Descriptive Statistics.

Panel A: Summary Statistics.

	N	Mean	S.D.	25%	Median	75%
Price Impact Variables:						
CAR[0,0]	416,266	−0.0003	0.080	−0.042	−0.002	0.040
CAR[1,4]	415,346	0.0003	0.045	−0.021	0.001	0.023
CAR[1,6]	415,426	0.0003	0.040	−0.017	0.002	0.021
Managerial Responses:						
DisAccrual	244,624	0.082	0.789	−0.168	0.102	0.399
Freq_Guidance	416,849	0.091	0.266	0.000	0.000	0.000
Bad News	47,370	0.540	0.498	0.000	1.000	1.000
Guidance Precision	47,370	0.174	0.379	0.000	0.000	0.000
Guidance Error	33,856	0.009	0.024	0.001	0.002	0.006
Main Independent Variables:						
MFFlow	416,849	−0.618	1.326	−0.619	−0.107	0.000
MFS	416,849	0.325	0.474	0.000	0.102	0.482
Analyst Coverage	416,849	0.968	0.966	0.000	0.693	1.792
Analyst Dispersion	106,625	0.158	0.463	0.017	0.035	0.093
Amihud	416,398	4.167	3.113	1.359	3.719	6.589
Bid-Ask Spread	415,751	0.027	0.047	0.002	0.012	0.033
ROA	416,849	−0.010	0.073	−0.007	0.004	0.017
Cash Flow	381,961	0.014	0.060	0.003	0.018	0.038
Operating Profit	347,175	0.026	0.052	0.006	0.028	0.050
Total InstOwn	416,713	0.430	0.319	0.130	0.396	0.703
Blockholder	416,713	1.661	1.609	0.000	1.000	3.000
Investor Turnover	404,419	0.253	0.065	0.212	0.247	0.285
Control Variables:						
Size	416,849	5.838	2.182	4.249	5.797	7.293
Tobin's Q	416,849	1.698	2.641	0.627	1.053	1.865
Leverage	416,849	0.214	0.233	0.026	0.160	0.330
Turnover	416,401	0.006	0.008	0.002	0.004	0.008
MRet	416,743	0.001	0.002	−0.000	0.001	0.002
BHAR	416,849	−0.006	0.528	−0.333	−0.070	0.206
Sigma	416,849	0.144	0.113	0.075	0.115	0.178
MF Ownership	416,755	0.117	0.121	0.013	0.077	0.191
Inst. Ownership	416,713	0.312	0.224	0.103	0.303	0.499
P_Surprise	373,291	0.372	0.483	0.000	0.000	1.000
N_Surprise	244,262	0.310	0.462	0.000	0.000	1.000
$\|Surprise\|$	244,262	0.007	0.018	0.000	0.001	0.005
Propmb	373,291	0.428	0.355	0.000	0.500	0.750
Guide_Prior	408,682	0.023	0.150	0.000	0.000	0.000

Panel B: Correlation Matrix.

Variables	$CAR[0,0]$	$CAR[1,4]$	*MFS*	*Guidance*	*DisAccrual*	*Analyst*	*Amihud*	*ROA*	*Total InstOwn*	*Size*
$CAR[0,0]$	1.000									
$CAR[1,4]$	0.054	1.000								
MFS	−0.013	0.056	1.000							
Guidance	0.013	0.026	0.078	1.000						
DisAccrual	0.047	0.036	0.028	0.016	1.000					
Analyst	0.017	0.040	0.168	0.291	0.023	1.000				
Amihud	0.002	−0.006	−0.237	−0.231	−0.027	−0.655	1.000			
ROA	0.104	0.117	0.140	0.077	0.172	0.160	−0.187	1.000		
Total InstOwn	0.033	0.071	0.382	0.256	0.029	0.546	−0.686	0.212	1.000	
Size	0.038	0.068	0.255	0.144	0.038	0.492	−0.719	0.356	0.563	1.000

This table presents the summary statistics and correlation matrix of the main variables used in our analyses. We winsorize all firm characteristics at the 1st and 99th percentiles. Definitions of the variables are provided in the Appendix.

e., quarter 0). The figure shows a negative abnormal return up to −3.8% (from −6 to +2 months) around fire sales which is reversed within one and half years. This is consistent with the idea that mutual fund sales due to extreme outflows result in stock underpricing. However, the price decline appears to have started in the three months leading up to the quarter of fire sales and persisted for another 2 months. Thus, we cannot completely rule out the possibility that mutual fund sales in quarter 0 may be motivated by negative firm fundamentals and/or that mutual fund fire sales might have been partially anticipated by the stock market.[10] Nevertheless, the fact that the price decline is eventually completely reversed suggests that mutual fund flow-driven sales are associated with some degree of

[10] Furthermore, the average stock price also continues to increase after the negative abnormal return is fully recovered.

stock underpricing.

We then examine whether mutual fund flow-induced fire sales have a significant price impact using the following model:

$$CAR_{i,t} = \beta_0 + \beta_1 MFS_{i,t} + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \tag{1}$$

where $Quarter_t$ and $Firm_i$ denote year-quarter and firm fixed effects, respectively, and $\varepsilon_{i,\ t}$ denotes the error term. We examine the average monthly cumulative abnormal return (*CAR*) around fire sales over various windows. We first use *CAR* in the quarter of fire sales (i.e., $CAR[0,0]$) to examine the initial price impact, and then use *CAR* over the subsequent two, four, and six quarters ($CAR[1,2]$, $CAR[1,4]$, $CAR[1,6]$) to examine the persistence of price shocks and the timing of reversals. $MFS_{i,\ t}$ is the natural logarithm of the absolute value of the magnitude of the shock: $MFS = Ln(1+|MFFlow|)$. We estimate standard errors robust to clustering at the firm level and heteroskedasticity. We control for lagged firm characteristics including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*.[11] We predict the coefficient for $MFS_{i,\ t}$ (β_1) to be negative for $CAR[0,0]$ if the fund flow-driven fire sales induce significant price drops.

Panel A of Table 2 shows the coefficient estimates for Eq. (1). Column (1) shows that there is a significantly negative abnormal return during the quarter of mutual fund fire sales. After controlling for firm characteristics and fixed effects (firm and year-quarter), a one standard deviation increase in *MFS* (0.474) is associated with a 1.14% negative abnormal return (0.38% per month) in that quarter.[12] This result shows that mutual funds' trading under extreme outflows produces a significantly negative price impact on the underlying stocks. The estimates in columns (2) and (3) suggest that there is an additional negative abnormal return in the four quarters *post* mutual fund fire sales. Column (4) shows that *MFS* is positively related to stock return in the six quarters after fire sales. Although this relation is statistically insignificant, it suggests a reversal in the stock price after mutual fund fire sales, consistent with the pattern presented in Fig. 1.

We also examine whether informational transparency helps mitigate the price impact of mutual fund fire sales using the following model:

$$\begin{aligned} CAR_{i,t} = {} & \beta_0 + \beta_1 MFS_{i,t} \times Info_High_{i,t-1} + \beta_2 MFS_{i,t} \times Info_Median_{i,t-1} \\ & + \beta_3 MFS_{i,t} \times Info_Low_{i,t-1} + \beta_4 Info_High_{i,t-1} + \beta_5 Info_Low_{i,t-1} \\ & + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{2}$$

We sort the sample into terciles based on proxies of information quality, and include the interaction terms between $MFS_{i,\ t}$ and binary variables indicating firms with proxies of information quality in the top/medium/bottom terciles in Eq. (2). We predict the negative price impact of mutual fund fire sales to decrease in magnitude across levels of informational transparency if firms with higher information quality in the market are less vulnerable to the non-fundamental price shocks. We use analysts' forecast quality and stock liquidity as proxies for informational transparency. Financial analysts are deemed to be important information producers in the market (Womack (1996); Kelly and Ljungqvist (2012)). Stock illiquidity reflects adverse selection cost arising from information asymmetry between market participants (Easley and O'Hara (2004)). Here we consider a firm as having better information quality if the stock has greater analyst coverage, lower analyst forecast dispersion, or higher stock liquidity as indicated by a lower value of *Amihud's* illiquidity or *Bid-Ask Spread*.

Panel B of Table 2 presents the estimates of Eq. (2). Columns (1) and (2) show that β_1, β_2, and β_3 are negative and decreasing, consistent with our prediction that firms with greater analyst coverage or lower analyst forecast dispersion tend to have less negative abnormal returns during mutual fund fire sales. Columns (3) and (4) report the estimates of Eq. (2) with stock liquidity as proxies of information quality. Consistent with the hypothesis that firms with liquid stocks are less affected by mutual fund price pressure, we find that the magnitude of the coefficients for *MFS* decreases with stock liquidity. This finding is also supported by Fig. 2a and b. In these figures, we first partition the sample into two groups according to the sample median of *Analyst Coverage* or *Amihud's Illiquidity*, then within each rgroup we plot the monthly cumulative average abnormal returns over the CRSP equal-weighted index from 6 months before to 18 months after mutual fund flow-driven fire sales. The figures show that firms with higher analyst coverage or liquidity are less affected by the negative price shocks. Overall, the results in Panel B of Table 2 and Fig. 2a and b are supportive of our hypothesis that a good information environment, where analysts have less divergence in earnings forecasts and stock liquidity is high, can help stabilize stock prices when firms experience exogenous market disruptions.[13]

4.2. Voluntary disclosure as a response to market disruptions

In this section, we test hypothesis *H*1 and examine whether firms respond to mutual fund fire sales by increasing voluntary disclosure. As discussed in the previous section, information transparency appears to mitigate the adverse effect of non-fundamental shocks to stock prices. It is plausible, therefore, that firms would want to enhance their informational transparency in response to such fire sale shocks. We focus on management earnings guidance and examine whether managers increase guidance frequency to improve

[11] A detailed description of the control variables is in section 3.4.

[12] Note that the constant term is included in our models, but it cannot be interpreted in a meaningful way when fixed effects are included in the regressions. For example, the estimated constant will depend on which year-quarter dummy is dropped from the regression.

[13] This evidence echoes prior studies by Roulstone (2003), Crawford et al. (2012), Kelly and Ljungqvist (2012), and Balakrishnan et al. (2014).

Table 2
The Price Impact of Mutual Fund Fire Sales.

Panel A: Price Impact over Various Horizons.

	(1)	(2)	(3)	(4)
Dependent Variable	$CAR[0,0]$	$CAR[1,2]$	$CAR[1,4]$	$CAR[1,6]$
MFS	−0.008***	−0.002***	−0.000	0.000
	(−26.21)	(−7.07)	(−1.10)	(1.63)
Size	−0.009***	−0.008***	−0.008***	−0.007***
	(−21.66)	(−19.33)	(−19.19)	(−17.69)
Leverage	−0.006***	−0.004***	−0.004***	−0.004***
	(−4.76)	(−3.01)	(−3.11)	(−3.71)
ROA	0.121***	0.016***	0.012***	0.013***
	(28.77)	(5.51)	(4.80)	(5.70)
Tobin's Q	−0.001***	−0.001***	−0.001***	−0.001***
	(−8.66)	(−9.33)	(−11.10)	(−10.82)
MF Ownership	0.010***	0.014***	0.013***	0.013***
	(3.94)	(6.21)	(6.30)	(6.24)
Inst. Ownership	0.002	0.002	0.003**	0.004***
	(1.61)	(1.12)	(2.52)	(3.34)
Sigma	0.013***	−0.007***	−0.007***	−0.005***
	(5.18)	(−3.31)	(−3.42)	(−2.89)
MRet	−2.374***	−0.563***	−0.645***	−0.847***
	(−12.75)	(−3.50)	(−4.61)	(−6.78)
BHAR	0.006***	−0.000	−0.001**	−0.001**
	(9.12)	(−0.82)	(−2.54)	(−2.18)
Turnover	−0.200***	−0.158***	−0.145***	−0.139***
	(−5.66)	(−5.13)	(−5.29)	(−5.38)
Analyst Coverage	0.000	−0.001***	−0.001***	−0.001***
	(0.46)	(−4.83)	(−5.24)	(−5.59)
Amihud	0.007***	0.006***	0.005***	0.004***
	(36.96)	(35.76)	(33.74)	(31.41)
Adjusted R^2	0.081	0.142	0.248	0.331
Observations	416,119	415,033	415,098	415,147
Firm & Year-Quarter Fixed Effects	YES	YES	YES	YES

Panel B: Information Environment and Price Impact.

Dependent Variable	$CAR[0,0]$			
	(1)	(2)	(3)	(4)
Information Measure	*Analyst Coverage*	*Analyst Dispersion*	*Amihud's Illiquidity*	*Bid-Ask Spread*
MFS × Info_High	−0.003***	−0.003***	0.000	0.000
	(−5.64)	(−3.30)	(0.90)	(0.95)
MFS × Info_Median	−0.009***	−0.003***	−0.009***	−0.010***
	(−19.52)	(−3.44)	(−20.81)	(−21.63)
MFS × Info_Low	−0.010***	−0.005***	−0.012***	−0.014***
	(−23.49)	(−4.98)	(−24.27)	(−24.20)
Info_High	−0.001**	0.001	−0.005***	−0.002***
	(−2.03)	(1.03)	(−7.13)	(−3.44)
Info_Low	−0.002***	0.002**	−0.001*	−0.002***
	(−3.16)	(2.45)	(−1.79)	(−3.92)
Adjusted R^2	0.081	0.099	0.081	0.082
Observations	416,119	106,603	416,119	415,472
P-value(*MFS*×High=*MFS*×Low)	0.000	0.036	0.000	0.000
Control Variables	YES	YES	YES	YES
Firm & Year-Quarter Fixed Effects	YES	YES	YES	YES

Panel A of this table presents estimates of the impact of mutual fund fire sales on stock returns over various horizons. The dependent variables are the average monthly abnormal returns during or after fire sales: the current quarter ($CAR[0,0]$), the subsequent two quarters ($CAR[1,2]$), the subsequent four quarters ($CAR[1,4]$), and the subsequent six quarters ($CAR[1,6]$). The independent variable of interest is *MFS*. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B presents estimates of the relation between the quality of information environment and the price impact of mutual fund fire sales. The dependent variable is the average monthly abnormal returns in quarter *t*, $CAR[0,0]$. The independent variables of interest include interactions between *MFS* of quarter *t* and binary variables indicating firms with high/median/low levels of information quality in $t-1$. We use four proxies to measure the quality of information environment: *Analyst Coverage*, *Analyst Dispersion*, *Amihud's Illiquidity*, and *Bid-Ask Spread*; the corresponding dummy variables (*Info_High*, *Info_Median*, and *Info_Low*) are based on terciles of each measure. We control for the same firm characteristics in quarter $t-1$ as in Panel A. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter

fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

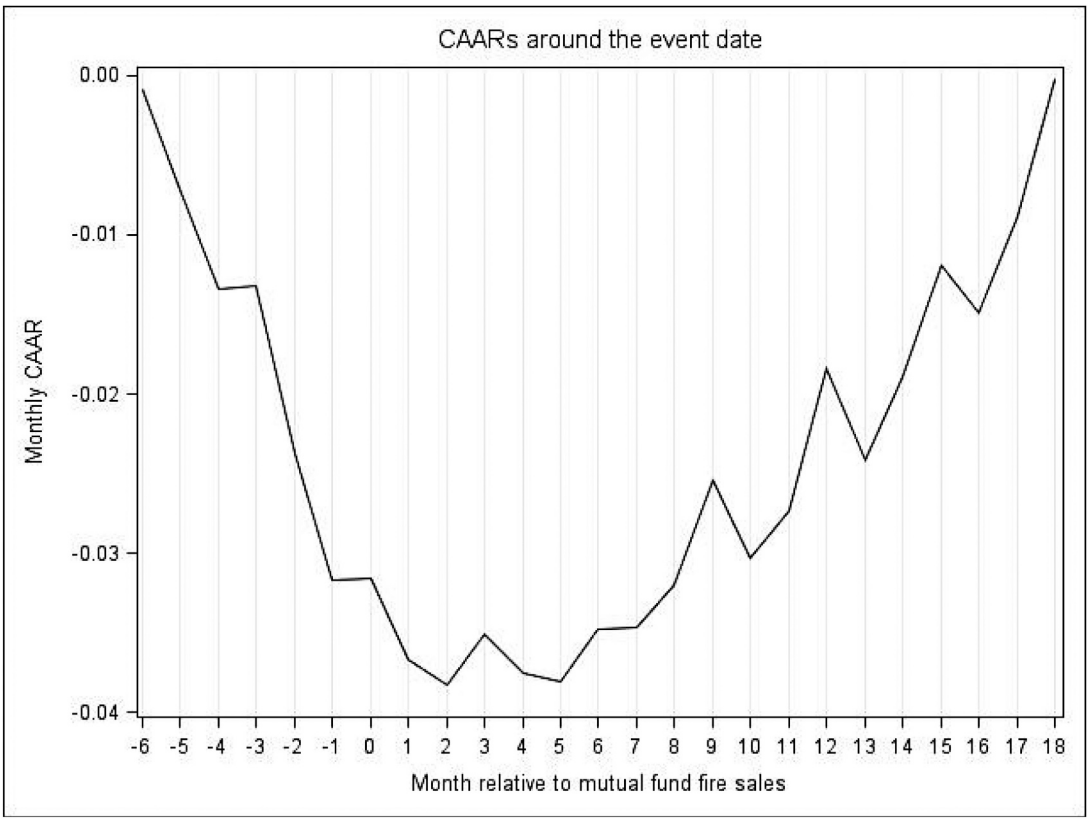


Fig. 1. This figure plots the monthly cumulative average abnormal returns over the CRSP equal-weighted index from 6 months before to 18 months after mutual fund flow-driven fire sales (*MFFlow* below the 10th percentile of the full sample). The fire sales occur from the beginning of month −2 to the end of month 0.

the information environment after mutual fund fire sales. The regression model is as follows:

$$Freq_Guidance_{i,t+1} \Big/ Guidance_D_{i,t+1} = \beta_0 + \beta_1 MFS_{i,t} + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \tag{3}$$

We estimate Eq. (3) in two ways. The first way is using OLS regressions where the dependent variable is $Freq_Guidance_{i,\ t+1}$, the natural logarithm of one plus the number of quarterly earnings forecasts made by the management in quarter $t+1$. Alternatively, we estimate the model using Logit regressions where the dependent variable is a dummy variable, $Guidance_D_{i,\ t+1}$, which equals one if a firm issues at least one quarterly earnings forecast over a fiscal quarter, and zero otherwise. In addition to firm characteristics used in the previous analysis, we also control for some determinants of management guidance documented by prior literature (e.g., Rogers et al. (2009); Rogers and Van Buskirk (2013)): *Guide_Prior* indicates whether the firm previously provided guidance for the earnings of the same quarter; *Surprise* (its absolute value is denoted as |*Surprise*|) is calculated as the absolute value of the reported actual earnings minus the most recent median analyst forecasts in quarter $t+1$, deflated by stock price three trading days prior to the earnings announcement date; *P_Surprise* is a dummy variable that indicates positive earnings surprises (*P_Surprise* equals one if *Surprise*$> +0.0001$); *N_Surprise* is a dummy variable that indicates negative earnings surprises (*N_Surprise* equals one if *Surprise*< -0.0001); *Propmb* is the proportion of the four prior quarters that have non-negative earnings surprises, i.e., meeting or beating analysts' expectations. We also control for firm and year-quarter fixed effects so that the coefficients capture within-firm variation in the frequency of management earnings guidance around mutual fund fire sales.

Table 3 presents the estimates of Eq. (3). In columns (1) and (2), we estimate OLS models where the dependent variable is

$Freq_Guidance_{i,\ t+1}$. Columns (3) and (4) report estimates of Logit regressions where the dependent variable is *Guidance_D*. The coefficient for *MFS* is significantly positive across all specifications, indicating that managers increase the frequency of earnings forecasts after mutual fund fire sales.[14] The coefficient magnitude suggests that the effect of mutual fund fire sales on guidance frequency is economically significant. For example, based on the estimate in column (2), a one standard increase in *MFS* is related to an increase of 2.1% in the frequency of earnings guidance from the sample mean.

We next look into the attributes of management earnings guidance in the presence of mutual fund fire sales. Earnings guidance can generally improve transparency and liquidity in the stock market (Coller and Yohn (1997); Balakrishnan et al. (2014)), though it is also possible that managers use upward-biased earnings guidance to counter the effect of mutual fund fire sales. Thus, it is important to know whether managers use earnings guidance to disclose useful information or convey false signals in response to negative price shocks.

We first check whether managers are more likely to issue good news in response to the negative price shocks. We classify each management earnings forecast as positive/negative relative to analysts' consensus, and create a dummy variable (*Bad News*) indicating whether a firm issues more negative forecasts than positive ones. We use a Logit model to estimate the effect of mutual fund fire sales on the likelihood of bad news issuance.[15] We present the estimates in column (1) of Table 4. The coefficient for *MFS* is positive but insignificant, suggesting that there may be as many positive forecasts as negative ones when fire sales occur. An interpretation of why firms might be willing to issue negative forecasts relative to analysts' consensus is that even negative forecasts could be useful in boosting price reversals, as long as the forecast is not as negative as what is implied by the depressed stock price. Since prior studies document that firms tend to issue more negative forecasts on average, our results are also consistent with firms adjusting their earnings forecasts upward in response to mutual fund fire sales.[16]

We also examine the effect of mutual fund fire sales on the precision of management earnings guidance. We estimate a Logit model where the dependent variable is a dummy variable that equals one if there are more point forecasts than range forecasts in the quarter subsequent to fire sales, and zero otherwise. The results are reported in column (2) of Table 4. We find a positive coefficient on *MFS*, indicating that firms increase the precision of forecasts in response to the fund flow-driven fire sales. Point forecasts are believed to convey more precise information and are associated with stronger market reactions than other types of forecasts (Baginski et al. (1993)).

Next, we test the effectiveness of management earnings forecasts in mitigating the price impact of mutual fund fire sales and facilitating price recovery (*Hypothesis* 4). We start by estimating the following model:

$$\begin{aligned} CAR[0,n]_{i,t} = {} & \beta_0 + \beta_1 MFS_{i,t} + \beta_2 MFS_{i,t} \times Managerial_Response_{i,t+1} + \beta_3 Managerial_Response_{i,t+1} \\ & + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{4}$$

The dependent variables are average monthly market-adjusted abnormal returns from the beginning of quarter t to four or six quarters after ($CAR[0,4]$ or $CAR[0,6]$), and the independent variable of interest is the interaction between mutual fund price pressure (*MFS*) and *Managerial_Response* in terms of earnings guidance. Here we define earnings guidance as a dummy variable, *Guidance_D*, that equals one if the firm management issues at least one quarterly earnings forecast in quarter $t+1$. Our hypothesis that post-fire-sales earnings guidance moderates the negative price impact of mutual fund fire sales predict a significantly positive coefficient estimate of β_2 in Eq. (4).

Regression results are presented in Panel A of Table 5. Column (1) shows that the coefficient for *MFS* × *Guidance_D* is significantly positive while that for *MFS* is negative. This suggests that mutual fund fire sales are associated with a negative price change from quarter t to quarter $t+4$, but that this price drop is weaker for guidance-issuing firms. In column (2), we also observe a weaker price drop from quarter t to quarter $t+6$ for guidance-issuing firms. Thus, the post-fire-sales earnings guidance is associated with a weaker price impact of mutual fund fire sales in the longer run.

However, the above model could raise a concern that firms that are willing to issue more earnings guidance are also the ones that are less vulnerable to stock-market disruptions in the first place. In other words, one may argue that the weaker price impact is not necessarily driven by the post-fire-sales earnings guidance, but related to a stock's inherently strong resilience to price shocks. To provide a sharper test of price recovery post fire sales, we examine the following model by focusing on firm-quarter observations that are in the bottom tercile of (non-zero) *MFFlow* in the quarter.[17]

$$\begin{aligned} CAR[1,n]_{i,t} = {} & \beta_0 + \beta_1 CAR[0,0]_{i,t} + \beta_2 CAR[0,0]_{i,t} \times Managerial_Response_{i,t+1} \\ & + \beta_3 Managerial_Response_{i,t+1} + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{5}$$

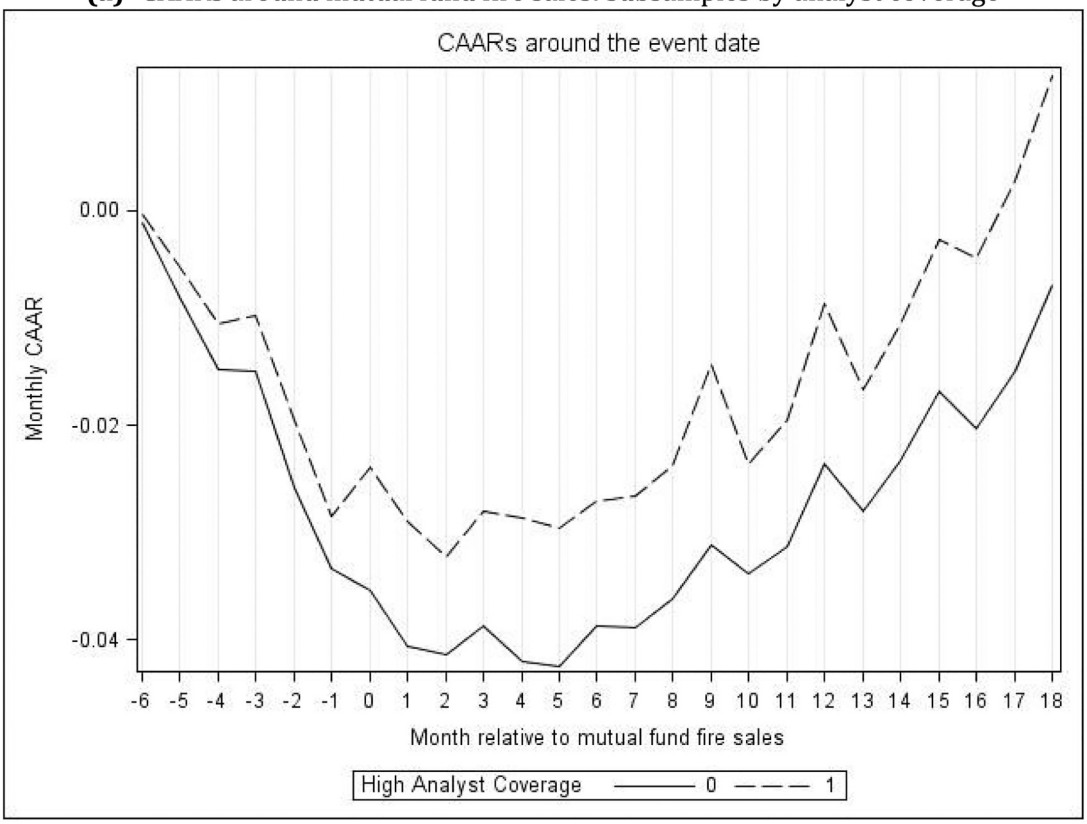
[14] The results hold if we estimate Poisson or negative binomial regressions for robustness.

[15] The sample size in this regression analysis is dramatically reduced since we only include quarterly point and range forecasts (i.e., close-end forecasts). Li and Zhang (2015) show that the calculation and classification of forecast news is more straightforward for these forecasts.

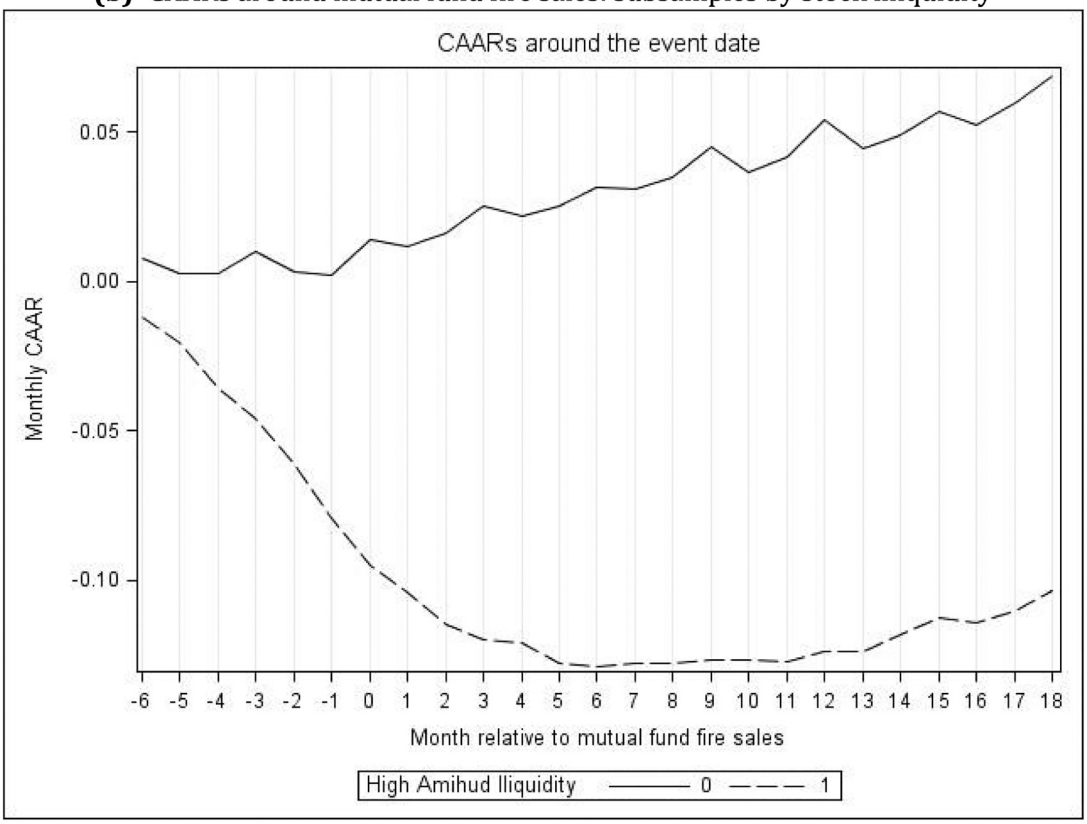
[16] See, e.g., Skinner (1994) finds that managers are willing to release relatively negative earnings forecasts, possibly to reduce the likelihood of shareholder lawsuits; Balakrishnan et al. (2014) document that in an average quarter, negative guidance is about twice as likely as positive guidance.

[17] In our full sample, about 63% of the firm-quarter observations have non-zero *MFFlow*. Thus, the bottom tercile of firm-quarter observations with non-zero *MFFlow* amount to 20% of the full sample.

(a) CAARs around mutual fund fire sales: subsamples by analyst coverage



(b) CAARs around mutual fund fire sales: subsamples by stock illiquidity



(caption on next page)

Fig. 2. These figures plot the subsamples, partitioned by the sample median of *Analyst Coverage* or *Amihud's Illiquidity*, monthly cumulative average abnormal returns over the CRSP equal-weighted index from 6 months before to 18 months after mutual fund flow-driven fire sales (*MFFlow* below the 10th percentile of the full sample). The fire sales occur from the beginning of month −2 to the end of month 0.

Table 3
Mutual Fund Fire Sales and Earnings Guidance.

Model	OLS	OLS	Logit	Logit
Dependent Variable	*Freq_Guidance*		*Guidance_D*	
	(1)	(2)	(3)	(4)
MFS	0.012***	0.004**	0.181***	0.082***
	(5.37)	(2.22)	(8.55)	(3.47)
Size		0.030***		0.412***
		(7.86)		(19.26)
Leverage		−0.012		−0.115
		(−1.31)		(−1.51)
ROA		0.055***		1.937***
		(3.33)		(8.09)
Tobin's Q		−0.000		−0.030***
		(−0.97)		(−5.35)
MF Ownership		0.074***		0.432***
		(4.06)		(3.42)
Inst. Ownership		−0.001		−0.032
		(−0.05)		(−0.42)
Sigma		−0.146***		−2.204***
		(−8.00)		(−13.54)
MRet		4.930***		34.902***
		(5.60)		(3.09)
BHAR		−0.019***		−0.161***
		(−6.10)		(−4.05)
Turnover		−0.561**		−13.795***
		(−2.53)		(−8.15)
Analyst Coverage		0.007***		0.086***
		(4.86)		(6.15)
Amihud		−0.004***		−0.159***
		(−2.79)		(−12.07)
P_Surprise		−0.011***		−0.096***
		(−5.13)		(−4.16)
N_Surprise		−0.025***		−0.302***
		(−10.55)		(−11.21)
|*Surprise*|		−0.148***		−3.842***
		(−3.89)		(−5.60)
Propmb		0.032***		0.468***
		(10.31)		(14.62)
Guide_Prior		0.500***		2.797***
		(36.44)		(63.93)
Adjusted R^2	0.390	0.449	0.083	0.156
Observations	244,201	244,201	160,790	160,790
Firm & Year-Quarter Fixed Effects	YES	YES	YES	YES

This table presents estimates of the effect of mutual fund fire sales on management earnings guidance. In columns (1) and (2), we estimate OLS models where the dependent variable is *Freq_Guidance*, the natural logarithm of one plus the frequency of quarterly earnings forecasts issued by firm management in quarter $t+1$. In columns (3) and (4), we estimate Logit models, and the dependent variable is *Guidance_D*, a dummy that equals one for firms issuing at least one earnings forecast in quarter $t+1$, and zero otherwise. The independent variable of interest for all the models is *MFS*. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We also control for key determinants of management guidance: *P_Surprise*, *N_Surprise*, |*Surprise*|, *Propmb*, and *Guide_Pior*, measured in quarter $t+1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

The idea is that since stock price starts recovering after the initial drop driven by fire sales in the subsequent quarters, the correlation between $CAR[1,n]$ and $CAR[0,0]$ should be negative (i.e., $\widehat{\beta}_1 < 0$). A significantly negative estimate for β_2 would then suggest that post-fire-sales earning guidance is associated with a stronger price reversal over the subsequent n quarters. We present the estimates in Panel B of Table 5. The estimates in column (1), where the dependent variable is *CAR* from one quarter to four quarters post fire sales, show that β_2 is −0.01 and significant at the 10% level. This coefficient estimate indicates that stock prices reverse more strongly within one year after mutual fund fire sales when managers issue more earnings forecasts as a response. Note that the predicted estimate for β_2 here should be negative, which is opposite to that of Eq. (4). This evidence is consistent with our hypothesis that earnings guidance helps mitigate the negative price impact of mutual fund flow-driven fire sales. In column (2) where we examine the average monthly abnormal returns over the subsequent six quarters, the coefficient for the interaction term is not significant. This is

likely because price recovery is largely over within one and half years whether or not earnings forecasts are issued. Taken together, a plausible interpretation for our results in columns (1) and (2) is that the price recovery post fire sales is more brisk in the presence of earnings guidance.

Columns (3) and (4) of Panel B of Table 5 show the estimates for the relation between management forecast error and price recovery. Forecast error is computed as the difference between the actual earnings and the management forecast (point estimate or the midpoint of a range forecast), scaled by beginning-of-quarter stock price. We create a dummy variable *Error_Low* that equals one if a forecast error is below the sample median, and zero otherwise. Again, we estimate Eq. (5) within the sample that is in the bottom tercile of *MFFlow* and has at least one quarterly earnings forecast in the subsequent quarter. The variable of interest is the interaction between $CAR[0,0]$ and *Error_Low*. The estimate for β_2 in columns (3) and (4) are -0.034 and -0.023, significant at the 5% and 10% level, respectively. This evidence suggests that higher quality forecasts, as indicated by a lower level of error, are related to faster price recovery.

Overall, our findings suggest that managers increase the issuance of earnings guidance in response to mutual fund fire sales, and this response appears useful in mitigating the adverse price effect of the mutual fund fire sales shock. These findings are thus supportive of hypothesis *H*4.

4.3. Earnings management as an alternative response to market disruptions

Despite the above evidence that earnings guidance appears effective in facilitating price recovery from mutual fund fire sales, it is not clear whether all firms, especially those with poorer performance, choose to respond in this fashion. In this section, we focus on underperforming firms to examine whether they respond to mutual fund fire sales by issuing earnings guidance or resort to another approach. We begin by modifying Eq. (3) into the following form:

$$\begin{aligned} Managerial_Response_{i,t+1} = \beta_0 &+ \beta_1 MFS_{i,t} \times Performance_High_{i,t-1} \\ &+ \beta_2 MFS_{i,t} \times Performance_Medium_{i,t-1} + \beta_3 MFS_{i,t} \times Performance_Low_{i,t-1} \\ &+ \beta_4 Performance_High_{i,t-1} + \beta_5 Performance_Low_{i,t-1} \\ &+ \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{6}$$

where we use three proxies of operating performance: *ROA*, *Cash Flow*, and *Operating Profit*. We sort the sample into terciles based on firms' historical performance, and interact *MFS* with binary variables indicating firms with high/medium/low levels of historical performance. Based on hypothesis *H*2, we predict that firms with worse historical performance are less likely to increase the frequency of earnings guidance in response to mutual fund fire sales. The notion is that, to the extent firm performance is persistent over time, firms with lower historical performance are also likely to underperform in the future. Thus, β_1, β_2, and β_3 should be decreasing in magnitude when the dependent variable is earnings guidance. The estimates reported in columns (1), (3), and (5) of Table 6 confirm our prediction: the coefficient for $MFS_{i,\ t}$ is insignificant when firm performance is low. Thus it appears that low performance deters firms from issuing voluntary earnings forecasts as a response to non-fundamental price shocks.

What might underperforming firms do when they expect that greater disclosure would not necessarily aid stock price recovery? We consider another potential response – earnings management. Earnings management is usually adopted to hide poor performance or postpone a portion of current good earnings to the future periods (Watts and Zimmerman (1978)). As we discuss in hypothesis *H*2, in the case of mutual fund fire sales, firms that have weak performance may respond to the fire sale by manipulating earnings upwards and sending false positive signals to the market. We therefore estimate Eq. (6) using earnings management as the dependent variable. We follow the literature and use discretionary accruals based on the modified Jones Model (Dechow et al. (1995)) as a proxy for earnings management. The estimates in columns (2), (4), and (6) of Table 6 show that, when using discretionary accruals as the dependent variable, the coefficient for *MFS* is significantly positive only for firms with lower levels of historical performance. Hence, consistent with *H*2, firms that have weaker historical operating performance appear to adopt income-increasing discretionary accruals as an alternative response to mutual fund price shocks.

We look further into the incentives of firms that manage earnings in response to mutual fund fire sales. As shown in the investment literature, mutual fund fire sales induce significant downward price pressure which is eventually corrected by the market (Coval and Stafford (2007)). Hence managers that focus on long-term fundamental value would not necessarily take any risky action to curb short-term mispricing. However, managers that are particularly concerned about current market valuation may be more proactive in responding to mutual fund fire sales. Extant literature shows that investors with short horizon can induce managerial myopia and suboptimal long-term investment decisions (Bushee (1998); Polk and Sapienza (2009)). In this case, managers that work for firms with higher ownership by short-term investors may be more concerned about restoring firm value in the short term. We thus test hypothesis *H*3, that firms that have short-term oriented shareholders are more (less) likely to use earnings management (voluntary disclosure) as a response to mutual fund price shocks. We estimate the following model:

$$\begin{aligned} Managerial_Response_{i,t+1} = \beta_0 &+ \beta_1 MFS_{i,t} \times Investor\ \ Horizon_{i,t-1} + \beta_2 MFS_{i,t} + \beta_3 Investor\ \ Horizon_{i,t-1} \\ &+ \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{7}$$

where we use three proxies for investor horizon: investors' turnover rate of their portfolios (Gaspar et al. (2005)), institutional ownership (Aghion et al. (2013)), and the number of blockholders (Edmans (2009)). The independent variables of interest are the

Table 4
Managerial Decisions on Forecast Characteristics.

Model	Logit	
	(1)	(2)
Dependent Variable	*Bad News*	*Forecast Precision*
MFS	−0.009	0.109**
	(−0.24)	(1.98)
Size	0.070**	−0.241***
	(1.99)	(−4.81)
Leverage	0.164	0.218
	(1.33)	(1.25)
ROA	1.535***	1.545***
	(4.24)	(2.89)
Tobin's Q	−0.007	0.016
	(−0.86)	(1.48)
MF Ownership	0.365*	0.471
	(1.92)	(1.63)
Inst. Ownership	0.103	0.300*
	(0.85)	(1.65)
Sigma	0.085	0.618*
	(0.34)	(1.68)
MRet	24.518	−29.033
	(1.39)	(−1.16)
BHAR	−0.191***	0.191**
	(−3.13)	(2.23)
Turnover	−1.276	10.160***
	(−0.52)	(2.87)
Analyst Coverage	0.052***	0.000
	(2.76)	(0.01)
Amihud	−0.230***	0.018
	(−10.39)	(0.56)
P_Surprise	−0.258***	−0.294***
	(−7.98)	(−6.63)
N_Surprise	0.160***	−0.385***
	(3.98)	(−6.77)
|*Surprise*|	2.130*	2.005
	(1.71)	(1.14)
Propmb	−0.641***	0.281***
	(−12.80)	(3.85)
Guide_Prior	−0.075	0.045
	(−1.60)	(0.72)
Adjusted R^2	0.040	0.069
Observations	39,587	31,425
Firm and Year-Quarter Fixed Effects	YES	YES

This table presents results from Logit regressions investigating the effect of mutual fund fire sales on management guidance attributes. The dependent variable in column (1) is *Bad News*, a binary variable that equals one if a firm has more bad news than good news in its management earnings forecasts in quarter $t + 1$, and zero otherwise. The dependent variable in column (2) is a binary variable, *Forecast Precision*, indicating whether a firm issues more point forecasts than range forecasts in quarter $t + 1$. Both models require firms to have at least one quarterly earnings forecast in $t + 1$. We control for firm characteristics and key determinants of a management guidance issuance, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, *Amihud*, *P_Surprise*, *N_Surprise*, |*Surprise*|, *Propmb*, and *Guide_Pior*. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

interactions between *MFS* and the proxies of investor horizon. Table 7 shows that the estimate on the interaction is significantly positive in column (2), where a higher *Investor Turnover* indicates shorter investor horizon. This result suggests that firms with more short-term oriented investors are more likely to increase discretionary accruals in response to mutual fund fire sales. The estimates in columns (4) and (6) also show that managers are less likely to engage in earnings manipulation when the firm's stocks are held by more institutional investors or blockholders.

As a comparison, we also estimate the above model using *Freq_Guidance* as the dependent variable. The estimates presented in columns (1), (3), and (5) show that, when there are relatively more long-term oriented investors, managers tend to choose voluntary disclosure as a response to non-fundamental price shocks. Therefore, our results are supportive of hypothesis *H*3: firm managers who may be particularly concerned about short-term price movements tend to use earnings management instead of information disclosure in response to mutual fund flow-driven fire sales.

Table 5
Earnings Guidance and the Price Impact of Mutual Fund Fire Sales.

Panel A: Earnings Guidance and Price Impact (Full Sample).

	(1)	(2)
Dependent Variable	$CAR[0,4]$	$CAR[0,6]$
MFS	−0.002***	−0.001***
	(−11.28)	(−7.36)
MFS × Guidance_D	0.001**	0.001**
	(2.40)	(2.21)
Guidance_D	−0.004***	−0.003***
	(−10.18)	(−8.69)
Adjusted R^2	0.284	0.357
Observations	408,023	408,041
Control Variables	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES

Panel B: Earnings Guidance and Price Recovery (Bottom Tercile of *MFFlow*).

	(1)	(2)	(3)	(4)
Dependent Variable	$CAR[1,4]$	$CAR[1,6]$	$CAR[1,4]$	$CAR[1,6]$
$CAR[0,0]$	−0.070***	−0.059***	−0.091***	−0.065***
	(−20.63)	(−21.48)	(−7.46)	(−6.41)
$CAR[0,0]$ × *Guidance_D*	−0.011*	−0.001		
	(−1.84)	(−0.27)		
$CAR[0,0]$ × *Error_Low*			−0.034**	−0.023*
			(−2.23)	(−1.91)
Guidance_D/Error_Low	−0.003***	−0.002***	−0.003***	−0.003***
	(−5.37)	(−4.48)	(−3.69)	(−4.37)
Adjusted R^2	0.324	0.396	0.356	0.438
Observations	92,403	92,406	10,009	10,009
Control Variables	YES	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES	YES

This table presents estimates on the relation between management earnings guidance and the price impact of mutual fund fire sales. In Panel A, the dependent variables are the average monthly abnormal returns from the event quarter to four quarters after ($CAR[0,4]$) and to six quarters after ($CAR[0,6]$). The independent variable of interest is the interaction between the magnitude of a fire sale shock (*MFS*) and a binary variable (*Guidance_D*) indicating whether a firm issues at least one earnings guidance in quarter $t+1$. The models of this panel cover the full sample and examine the differential price impacts of fire sales, depending on a firm's disclosure policy on issuing earnings guidance. In Panel B, we only include firm-quarter observations that experience extreme mutual fund fire sales (i.e., non-zero *MFFlow* in the bottom tercile), and examine the effect of issuing earnings guidance on the speed of price recovery after fire sales. The dependent variables are the average monthly abnormal returns over the four quarters ($CAR[1,4]$) and six quarters ($CAR[1,6]$) subsequent to fire sales. In columns (1) and (2), the independent variable of interest is the interaction between event-quarter average monthly abnormal returns ($CAR[0,0]$) and a binary variable (*Guidance_D*) indicating whether a firm provides at least one earnings guidance in quarter $t+1$. In columns (3) and (4), the independent variable of interest is the interaction between $CAR[0,0]$ and a binary variable indicating whether forecast error is below the sample median (*Error_Low*). We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics (in parentheses) are computed using robust, firm-clustered standard errors. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Although we observe both the use of earnings guidance and earnings management in response to mutual fund fire sales, these two actions have opposite implications for the information environments: earnings guidance improves transparency while earnings management induces opacity. As our analysis shows, informational transparency rather than opacity is more likely to help correct mispricing. Further, our findings suggest that firms that use earnings management are associated with lower operating performance. Hence, we expect earnings management to be associated with slower price recovery because of opacity as well as lower operating performance. We test this prediction (*H*4) by repeating the analysis as in Eqs. (4) and (5) for discretionary accruals.

In Panel A of Table 8 we present the estimates of Eq. (4) for the effect of discretionary accruals on the overall price impact of fire sales. We interact *MFS* with a dummy variable (*DisAccrual_High*) that equals one if the firm's discretionary accrual is above the sample median in quarter $t+1$. Consistent with hypothesis *H*4, we do not find that post-fire-sales discretionary accruals mitigate the negative price impact of the fire sales; instead, the negative price impact appears stronger when discretionary accruals are higher, as the coefficient for the interaction term is significantly negative in both columns. We also estimate the degree of price reversal using Eq. (5)

Table 6
Firm Performance and Managerial Response to Mutual Fund Fire Sales.

Performance Measure	ROA		Cash Flow		Operating Profit	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	*Freq_Guidance*	*DisAccrual*	*Freq_Guidance*	*DisAccrual*	*Freq_Guidance*	*DisAccrual*
MFS × *Performance_High*	0.007**	−0.009**	0.012***	−0.005	0.009***	−0.007
	(2.54)	(−2.21)	(3.88)	(−1.05)	(2.66)	(−1.54)
MFS × *Performance_Median*	0.002	−0.000	0.001	−0.003	0.007**	0.002
	(0.79)	(−0.11)	(0.54)	(−0.61)	(2.15)	(0.40)
MFS × *Performance_Low*	0.002	0.047***	−0.003	0.061***	−0.005*	0.063***
	(0.66)	(7.07)	(−1.27)	(6.78)	(−1.81)	(6.67)
Performance_High	0.002	0.052***	0.002	0.065***	0.003	0.087***
	(0.89)	(10.18)	(0.51)	(10.93)	(1.01)	(15.36)
Performance_Low	−0.014***	−0.068***	−0.020***	−0.064***	−0.020***	−0.151***
	(−5.11)	(−10.01)	(−7.26)	(−7.39)	(−6.49)	(−17.61)
Adjusted R^2	0.450	0.316	0.459	0.316	0.464	0.347
Observations	244,201	241,251	225,321	224,170	201,952	202,259
P-value(MF_D*ROA_High = MF_D*ROA_Low)	0.216	0.000	0.000	0.000	0.001	0.000
Control Variables	YES	YES	YES	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES	YES	YES	YES

This table presents estimates of the relation between historical firm performance and managerial response to fire sales. The dependent variables are *Freq_Guidance* and *DisAccrual* in quarter $t + 1$, and the independent variables of interest are the interactions between *MFS* and binary variables indicating a firm's lagged performance in the top/medium/bottom tercile. Three proxies are employed to measure firm performance: *Return on Assets (ROA)*, *Cash Flow*, and *Operating Profit*. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t - 1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

among firm-quarter observations that have extreme mutual fund outflows, i.e., firm-quarter observations that are in the bottom tercile of (non-zero) *MFFlow* in the quarter. The estimates in Panel B show that discretionary accruals are also not related to a stronger price reversal in either four quarters or six quarters following the fire sales. Hence, earnings management does not seem to mitigate the price impact of fire sales or facilitate price recovery.[18]

4.4. Effectiveness of managerial responses: Addressing Endogeneity concerns

Our empirical analysis indicates that firms endogenously choose between earnings guidance and earnings management as a response to mutual fund fire sales. This raises concerns about our interpretation of the price-recovery analysis. For instance, since firms' disclosure responses are related to firm performance, it is possible that better-performing firms would have had a faster recovery from fire sales regardless of their disclosure policies. Hence, we attempt to address the endogeneity concerns by exploiting exogenous changes in disclosure policies due to regulatory changes. Specifically, we look into the passage of the Regulation Fair Disclosure of 2000 (Regulation FD) and the Sarbanes-Oxley Act of 2002 (SOX). Regulation FD mandates that all public firms must disclose material information to all investors at the same time. Evidence shows that there is a substantial increase in the volume of voluntary earnings forecasts after the implementation of Regulation FD (Heflin et al. (2003)). SOX was enacted in response to a number of accounting scandals such as those of Enron and WorldCom and was intended to enhance shareholder protection from accounting frauds. Prior studies show that the passage of SOX caused a substantial decrease in the use of discretionary accruals (Cohen et al., 2008) and enhanced corporate transparency due to an increase in board independence (Armstrong et al. (2014)). Therefore, we use the passage of SOX to identify exogenous changes in the use of accrual-based earnings management during mutual fund fire sales. As we discuss in hypothesis *H*5, firms shift from earnings management to voluntary disclosure as a response to mutual fund fire sales due to Regulation FD/SOX may facilitate price recovery. We first test whether Regulation FD/SOX causes a shift in firms' response to exogenous price shocks using the following difference-in-differences model:

$$\begin{aligned} Managerial\ \ Response_{i,t+1} = {} & \beta_0 + \beta_1 MFS_{i,t} + \beta_2 MFS_{i,t} \times \mathrm{Post}_t + \beta_5 \mathrm{Post}_t \\ & + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}, \end{aligned} \tag{8}$$

$Post_t$ is a binary variable that equals one for firm-quarter observations after the effective date of Regulation FD or SOX. The dependent variable is earnings guidance (discretionary accruals) when the regulatory change is Regulation FD (SOX). Our prediction is that if Regulation FD (SOX) causes managers to increase (decrease) earnings guidance (discretionary accruals) as a response to mutual

[18] There are two possible explanations for why firms engage in earnings management even if it does not appear to benefit price recovery. The first possibility is that they falsely expect this action to be effective in reversing the stock price. The second possibility is that, if investors expect firms to engage in earnings management in equilibrium, they might rationally discount reported earnings for some level of anticipated manipulation. In this case, a firm that did not manipulate its earnings might be regarded as being worse than it actually was.

fund fire sales, then β_2 should be significantly positive (negative) for the model of earnings guidance (discretionary accruals). Column (1) in Panel A (B) of Table 9 presents the estimates of the above model with earnings guidance (discretionary accruals) as the dependent variable. We find that the estimate of β_2 is significantly positive for earnings guidance but negative for discretionary accruals. This evidence suggests that, after the passage of Regulation FD (SOX), firms increase (reduce) the use of earnings guidance (earnings management) in response to mutual fund fire sales. Thus, both natural experiments allow us to distinguish the effect of managerial response on price stability from firm attributes and managerial incentives. To further examine whether these exogenous changes in the choice of managerial response is related to a different degree of price recovery, we estimate the following difference-in-differences model among observations with (non-zero) *MFFlow* in the bottom tercile:

$$CAR[1,n]_{i,t} = \beta_0 + \beta_1 CAR[0,0]_{i,t} + \beta_2 CAR[0,0]_{i,t} \times Post_t + \beta_3 Post_t + \sum_k \beta_k Controls_{i,t-1} + Quarter_t + Firm_i + \varepsilon_{i,t}. \tag{9}$$

Similar to our prediction for Eq. (5), if the increases (decreases) in earnings forecasts (discretionary accruals) due to Regulation FD (SOX) contribute to faster price recoveries after mutual fund fire sales, we should expect β_2 to be significantly negative. In column (2) of Panel A and column (1) of Panel C, we show that the implementations of both regulations are associated with stronger price reversals in the four quarters subsequent to fire sales. This is consistent with the prediction that higher transparency as a result of expanded voluntary disclosure and less accounting manipulation, accelerates the price recovery process.

To further strengthen the causal interpretation of the result, we examine cases where firms are more/less likely to be affected by SOX. Some firms were already compliant with the key requirements of SOX prior to the passage of the new regulation. These firms provide a good counterfactual for the effect of disclosure policies on price recovery because their disclosure policies are unlikely to change due to the new regulation. We further partition the fire-sales sample into two subsamples, in terms of whether or not firms were compliant with the key requirements of SOX (i.e., having a majority of independent directors and a fully independent audit committee) prior to its passage, and apply the above models to examine whether SOX has differential effects on disclosure policies and post-fire-sales price recovery in the two subsamples. We determine the status of SOX compliance for S&P 1500 firms from 1998 to 2001, and perform diff-in-diff tests for the compliant and non-compliant samples. As indicated, in columns (2) and (3) of Panel B, we find that only the non-compliant firms reduce the use of discretionary accruals in response to mutual fund fire sales after the passage of SOX.

Table 7
Investor Horizon and Managerial Response to Mutual Fund Fire Sales.

Investor Horizon Measure	Investor Turnover		Inst. Ownership		Blockholder	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	*Freq_Guidance*	*DisAccrual*	*Freq_Guidance*	*DisAccrual*	*Freq_Guidance*	*DisAccrual*
MFS	0.018**	−0.055***	−0.011**	0.028***	−0.005*	0.025***
	(2.16)	(−3.68)	(−2.48)	(4.38)	(−1.74)	(4.73)
MFS × *Investor Turnover*	−0.056*	0.262***				
	(−1.73)	(4.25)				
MFS × *Total InstOwn*			0.026***	−0.040***		
			(3.37)	(−3.56)		
MFS × *Blockholder*					0.004***	−0.006***
					(3.32)	(−3.84)
Investor Turnover	−0.141***	0.051				
	(−5.84)	(0.85)				
Total InstOwn			0.267*	0.366		
			(1.67)	(1.17)		
Blockholder					−0.007***	−0.005**
					(−5.68)	(−2.57)
Adjusted R^2	0.450	0.318	0.449	0.315	0.450	0.315
Observations	241,589	233,921	244,201	241,251	244,201	241,251
Control Variables	YES	YES	YES	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES	YES	YES	YES

This table presents estimates of the relation between investor horizon and managerial response to fire sales. The dependent variables are *Freq_Guidance* and *DisAccrual* in quarter $t+1$. The independent variables of interest include the interactions between *MFS* and three proxies for investor horizon: *Investor Turnover*, *Inst. Ownership*, and *Blockholder*. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics with firm-clustered standard errors are in parentheses. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Moreover, the association between the passage of SOX and faster price recovery only holds for the non-compliant firms, as revealed in column (5) of Panel C. These results suggest a causal relation between disclosure policy changes and price recovery from market disruptions.

Table 8
Earnings Management and the Price Impact of Mutual Fund Fire Sales.

Panel A: Earnings Management and Price Impact (Full Sample)		
	(1)	(2)
Dependent Variable	$CAR[0,4]$	$CAR[0,6]$
MFS	−0.002***	−0.001***
	(−7.96)	(−4.84)
MFS × *DisAccruals_High*	−0.001**	−0.001**
	(−2.38)	(−2.18)
DisAccruals_High	−0.000	−0.000*
	(−0.70)	(−1.95)
Adjusted R^2	0.302	0.380
Observations	244,205	244,215
Control Variables	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES
Panel B: Earnings Management and Price Recovery (Bottom Tercile Fire Sales Sample)		
	(1)	(2)
Dependent Variable	$CAR[1,4]$	$CAR[1,6]$
$CAR[0,0]$	−0.071***	−0.061***
	(−14.78)	(−15.50)
$CAR[0,0]$ × *DisAccruals_High*	−0.006	−0.005
	(−0.97)	(−0.89)
DisAccruals_High	−0.001***	−0.001***
	(−4.46)	(−4.87)
Adjusted R^2	0.347	0.427
Observations	54,621	54,624
Control Variables	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES

This table presents estimates on the relation between earnings management and the price impact of mutual fund fire sales. In Panel A, we use the full sample and examine the differential effects of fire sales on stock prices, depending on a firm's level of discretionary accruals. The dependent variables are the average monthly abnormal returns from the event quarter to four quarters after ($CAR[0,4]$) and to six quarters after ($CAR[0,6]$). The independent variable of interest is the interaction between *MFS* and a binary variable that equals one if $DisAccrual_{t+1}$ above the sample median. In Panel B, we only include firm-quarter observations that experience extreme mutual fund fire sales (non-zero *MFFlow* in the bottom tercile), and examine the effect of earnings management on the speed of price recovery after fire sales. The dependent variables are the average monthly abnormal returns four quarters ($CAR[1,4]$) and six quarters ($CAR[1,6]$) subsequent to fire sales. The independent variable of interest is the interaction between event-quarter average monthly abnormal returns ($CAR[0,0]$) and a binary variable indicating a firm having $DisAccrual_{t+1}$ above the sample median. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We winsorize the variables at the 1*st* and 99*th* percentiles. We also include firm fixed effects and fiscal and calendar year-quarter fixed effects in the regressions. *t*-statistics (in parentheses) are computed using robust, firm-clustered standard errors. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

5. Conclusion

We examine managers' decisions regarding voluntary disclosure and financial reporting when there is a large price drop due to market disruptions. Using the mutual fund flow-driven fire sales as an exogenous market disruption that causes price declines, we find that previously well-performing firms increase the frequency of earnings guidance, while previously underperforming firms engage in income-increasing accrual-based earnings management. Further investigation shows that the choice of earnings management is driven by managers' concerns over short-term price levels and their relatively poor performance. By comparing the two types of responses, we show that earnings guidance is more effective than earnings management in mitigating the adverse price impact of mutual fund fire sales. To address endogeneity issues, we examine the effects of Regulation FD and SOX. In particular, we show that after the passage of SOX, firms that were previously not compliant with SOX's requirements were less likely to respond to fire sales by engaging in earnings management. These firms also appear to recover faster from mutual fund fire sales after the passage of SOX. This suggests a causal relation between disclosure policy changes and price recovery from market disruptions. Overall, our findings provide insight into firms' strategies in responding to disruptions in secondary stock markets, and highlight the importance of informational transparency in stabilizing prices during such disruptions.

Appendix A. Variable Definitions

A.1. Dependent variables

- *CAR* is average monthly abnormal returns over various windows, where the abnormal return is computed as a firm's monthly raw return minus the return on the CRSP value-weighted index.

Table 9
The regulation fair disclosure, Sarbanes-Oxley Act and managerial response to mutual fund fire sales.

Panel A: The Impact of Regulation FD on Earnings Guidance and Price Recovery.

Sample	*Full*	*Fire Sale*	Fire Sale
	(1)	(2)	(3)
Dependent Variable	*Guidance_D*	*CAR*[1, 4]	*CAR*[1, 6]
MFS	−0.000		
	(−0.00)		
MFS × *Post_RegFD*	0.118**		
	(2.41)		
CAR[0, 0] × *Post_RegFD*		−0.015**	0.001
		(−2.55)	(0.26)
CAR[0, 0]		−0.061***	−0.060***
		(−11.66)	(−13.80)
Post_RegFD	0.085	−0.004	−0.002
	(0.50)	(−1.56)	(−0.90)
Adjusted R^2	0.156	0.324	0.396
Observations	160,790	92,403	92,406
Control Variables	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES

Panel B: The Impact of SOX on Earnings Management

Dependent Variable	*DisAccrual*		
	(1)	(2)	(3)
Sample	*Full*	*Compliant*	*Non-compliant*
MFS	0.019***	0.005	0.020**
	(4.37)	(0.38)	(2.50)
MFS × *Post_SOX*	−0.020***	−0.014	−0.050***
	(−2.97)	(−0.75)	(−4.34)
Post_SOX	0.083*	0.086*	0.007
	(1.71)	(1.68)	(0.09)
Adjusted R^2	0.315	0.429	0.342
Observations	241,251	24,987	77,205
Control Variables	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES

Panel C: The Effect of SOX on Price Recovery

Sample	*Full (Fire Sale)*		*Compliant*		*Non-compliant*	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	*CAR*[1, 4]	*CAR*[1, 6]	*CAR*[1, 4]	*CAR*[1, 6]	*CAR*[1, 4]	*CAR*[1, 6]
CAR[0, 0]	−0.054***	−0.056***	−0.067***	−0.063***	−0.030***	−0.040***
	(−13.03)	(−16.45)	(−7.50)	(−9.93)	(−4.49)	(−7.13)
CAR[0, 0] × *Post_SOX*	−0.010*	0.004	0.013	0.019*	−0.026***	−0.006
	(−1.94)	(0.92)	(0.98)	(1.94)	(−3.00)	(−0.78)
Post_SOX	−0.001	−0.001	−0.006	−0.004	−0.008	−0.000
	(−0.17)	(−0.28)	(−0.94)	(−0.96)	(−1.59)	(−0.06)
Adjusted R^2	0.318	0.394	0.289	0.348	0.253	0.319
Observations	92,403	92,406	10,627	10,627	38,317	38,317
Control Variables	YES	YES	YES	YES	YES	YES
Firm and Year-Quarter Fixed Effects	YES	YES	YES	YES	YES	YES

Panel A presents estimates for the effect of Regulation FD on management guidance policies and price recovery post fire sales. The dependent variable is *Guidance_D* measured in quarter $t+1$ in column (1) and *CAR* over different event windows in columns (2) and (3). Column (1) covers the full sample, and columns (2) and (3) include firm-quarter observations experiencing extreme mutual fund fire sales (non-zero *MFFlow* in the bottom tercile). The independent variable of interest is the interaction between *MFS* and *Post_RegFD* in column (1), and that is the interaction between *CAR* [0, 0] and *Post_RegFD* in columns (2) and (3). Panel B presents estimates for the effect of SOX on earnings management post fire sales. The dependent variable is *DisAccrual*. The independent variable of interest is the interaction between *MFS* and *Post_SOX*. Column (1) covers the full sample, and columns (2) and (3) include observations that were compliant or not compliant with the key requirements of SOX prior to its passage, respectively. Panel C presents estimates for the effect of SOX on price recovery post fire sales. The dependent variable is *CAR* over different event windows. The independent variable of interest is the interaction between *CAR*[0, 0] and *Post_SOX*. The sample of column (1) includes observations under extreme fire sales (non-zero *MFFlow* in the bottom tercile). Among this extreme fire sales sample, we further partition firm-quarter observations into compliant or non-compliant groups based on the key requirements of SOX prior to its passage, and the results are presented in columns (2) and (3), respectively. We control for firm characteristics, including *Size*, *Leverage*, *ROA*, *Tobin's Q*, *MF Ownership*, *Inst. Ownership*, *Sigma*, *MRet*, *BHAR*, *Turnover*, *Analyst Coverage*, and *Amihud*, measured in quarter $t-1$. We also include firm fixed effects and year-quarter fixed effects in the regressions. *t*-statistics (in parentheses) are computed using robust, firm-clustered standard errors. *, ** and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

- *DisAccrual* is the value of discretionary accruals, where the discretionary accruals are measured using modified Jones Model (Dechow et al. (1995)). Its corresponding binary variable is *DisAccrual_D*, that equals one if the firm's discretionary accrual is above the sample median over a fiscal quarter, and zero otherwise.
- *Freq_Guidance* is the natural logarithm of one plus the number of quarterly earnings forecasts provided by a firm's management over a fiscal quarter.
- *Guidance_D* is a binary variable that equals one if a firm's management team issues at least one quarterly earnings forecast over a fiscal quarter, and zero otherwise.
- *Bad News* is a binary variable that equals one if a firm issues more negative forecast news (*Forecast News*) than positive news. *Forecast News* is the difference between a management earnings forecast (i.e., point estimate, or the midpoint estimate of a range forecast) and the consensus analyst forecast, scaled by beginning-of-quarter stock price. According to the sign of forecast news, each earnings forecast is classified as conveying positive forecast news ($ForecastNews > 0$) or negative forecast news ($ForecastNews < 0$). Consensus analyst forecast is the median of analyst forecasts issued at the time of a management earnings forecast.
- *Guidance Precision* (*PRECISION*) is a binary variable that equals one if a firm issues more point forecasts than range forecasts in a fiscal quarter, and zero otherwise.
- *Guidance Error* (*ERROR*) is defined as the absolute difference between management forecast estimate and the actual earnings, scaled by beginning-of-quarter stock price.

A.2. Main Independent Variables

- *MFFlow* is the hypothetical mutual fund outflow, measuring the magnitude of a non-fundamental shock to stock prices. See Section 3.3 for a detailed explanation of the construction of *MFFlow*.
- *MFS* is the natural logarithm of one plus the absolute value of *MFFlow*: $MFS = Ln(1+|MFFlow|)$.
- *Analyst Coverage (Analyst)* is the average number of analysts following the stock for the fiscal quarter. It is coded as 0 if there is no coverage from I/B/E/S.
- *Analyst Dispersion* is defined as standard deviation of analyst earnings forecasts during a quarter, scaled by beginning-of-quarter share price.
- *Amihud's Illiquidity* (*Amihud*) is defined as $ln(1 + AvgILLIQ \times 10^6)$, where *AvgILLIQ* is an average illiquidity for stock *i* in quarter *t* from daily data. Specifically, *AvgILLIQ* is calculated as the absolute return divided by dollar trading volume: $AvgILLIQ_{i,t} = \frac{1}{Days_{i,t}}\sum_{d=1}^{Days_{i,t}}\frac{|R_{i,t,d}|}{DolVol_{i,t,d}}$, where $Days_{i,\,t}$ is the number of valid observation days for stock *i* in fiscal year *t*, and $R_{i,\,t,\,d}$ and $DolVol_{i,\,t,\,d}$ are the return and dollar trading volume of stock *i* on day *d* in the current quarter *t*.
- $Bid-Ask\ Spread_{i,t} = \frac{1}{Days_{i,t}}\sum_{d=1}^{Days_{i,t}}\frac{Ask_{i,t,d}-Bid_{i,t,d}}{(Ask_{i,t,d}+Bid_{i,t,d})/2}$ where $Days_{i,\,t}$ is the number of observations for stock *i* in fiscal year *t*, and $Ask_{i,\,t,\,d}$ and $Bid_{i,\,t,\,d}$ are the closing ask and bid prices of the stock *i* on day *d* of quarter *t*.
- *Return on Assets (ROA)* is earnings before extraordinary items divided by lagged total asset.
- *Cash Flow* is earnings before extraordinary items plus depreciation expenses and R&D expenditure, divided by lagged total asset.
- *Operating Profit* is EBITDA plus R&D expenditure divided by lagged total asset.
- *Blockholder* is the number of institutional investors that hold more than 5% of a firm's stock.
- *Total InstOwn* is the ratio of the number of shares held by institutional investors to the total number of shares outstanding for a firm.
- *Investor Turnover* is measured based on the average of investors' churn rate, following Gaspar et al. (2005).
- *Regulation FD* is an indicator that equals one if quarter *t* is post October 2000, and zero otherwise.
- *SOX* is an indicator that equals one if quarter *t* is post July 2002, and zero otherwise.

A.3. Control Variables

- *Size* is the natural logarithm of total assets at the end of last quarter.
- *Tobin's Q* is computed as market value of equity plus book value of assets minus book value of equity, scaled by book value of assets.
- *Sigma* is standard deviation of a firm's monthly stock returns over the year prior to the current quarter.
- *Turnover* is calculated as the mean of daily trading volume divided by the total number of shares outstanding over the current quarter.
- *MRet* is the mean of daily returns over the current quarter.
- *BHAR* is buy-and-hold abnormal returns over the year prior to the current quarter. It is calculated as a firm's geometrically-compounded monthly returns minus the corresponding returns on the CRSP value-weighted index.
- *Leverage* is the sum of long-term debt and debt in current liabilities divided by total assets.
- *Inst. Ownership* is the ratio of the number of shares held by institutional investors (excluding mutual funds in the calculation of *MFFlow*) to the total number of shares outstanding for a firm.
- *MF Ownership* is the ratio of the number of shares held by mutual fund investors (included in the calculation of *MFFlow*) to the total number of shares outstanding for a firm.
- *P_Surprise* is a binary variable that equals 1 if this quarter's earnings surprise exceeds +0.0001.
- *N_Surprise* is a binary variable that equals 1 if this quarter's earnings surprise falls below −0.0001.
- |*Surprise*| is the absolute value of the actual earnings minus analyst consensus forecast, deflated by stock price.

- *Propmb* is the proportion of the previous 4 quarters that the firm's reported earnings met or beat analyst consensus forecast.
- *Guide_Prior* is a binary variable that equals 1 if the firm previously provided earnings guidance for the current quarter's earnings.

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